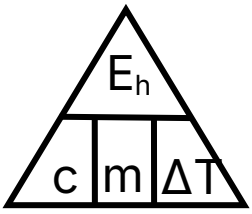
Calculating Heat Energy, E_h

The formula for calculating heat (or thermal) energy is:



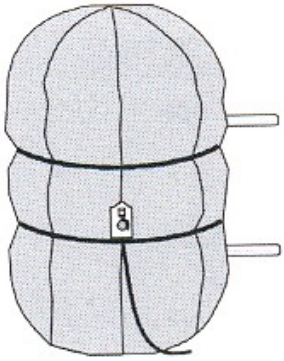
E_h= cmΔT

Heat Energy = specific heat capacity of a material x mass x change in temperature

- **Energy** is measured in **Joules** (**J**)
- **Specific heat capacity** is measured in **$Jkg^{-1}K^{-1}$** (kJ/kgK) and is the amount of energy required to raise the temperature of 1kg of the material by 1°C
- **Mass** is measured in **kg**
- **Change in temperature** (**ΔT**)('delta T') is measured in **Celsius** (**°C**) or Kelvin (**K**)

Example

A hot water tank contains 200 litres of water at 18°C. Calculate how much energy is required to raise the temperature of the water to 50°C. 1 litre of water has a mass of 1kg. The specific heat capacity of water is 4180 $Jkg^{-1}K^{-1}$. The specific heat capacity of a liquid changes depending on the type of liquid. The provided data booklet will specify what value to use.



E_h= cmΔT

= 200 x 4180 x 32

= 26,752,000 J

= 26.752 MJ

= **27 MJ** to two sig. fig.

What are my knowns?
m = 200kg
c = 4180 $Jkg^{-1}K^{-1}$
$\Delta T = t_2 - t_1 = 50 - 18 =$ 32°C

Questions

(Calculate answer to two significant figures)

1. Using a kettle with 1.5 Litre of water and a thermometer, work out the heat energy needed to raise the temperature from 5°C to:

   a) 50°C

   $\Delta T = t_2 - t_1 = 50 - 5 = 45°C$

   $c = 4180 Jkg^{-1}K^{-1}$

   $m = 1.5kg$

   $E_h = cm\Delta T$

   $E_h = 4180 \times 1.5 \times 45$

   $E_h = 282{,}150J = 282.15kJ$

   $E_h =$ **280 kJ** to two significant figures

   b) 40°C

   $\Delta T = t_2 - t_1 = 40 - 5 = 35°C$

   $c = 4180 Jkg^{-1}K^{-1}$

   $m = 1.5kg$

   $E_h = cm\Delta T$

   $E_h = 4180 \times 1.5 \times 35$

   $E_h = 219{,}450\ J = 219.45kJ$

   $E_h =$ **220 kJ** to two significant figures

c) 80°C

$\Delta T = t_2 - t_1 = 80 - 5 = 75°C$

$c = 4180 Jkg^{-1}K^{-1}$

$m = 1.5kg$

$E_h = cm\Delta T$

$E_h = 4180 \times 1.5 \times 75$

$E_h = 470{,}250\ J = 470.25kJ$

$E_h =$ **470 kJ** to two significant figures

d) 100°C

$\Delta T = t_2 - t_1 = 100 - 5 = 95°C$

$c = 4180 Jkg^{-1}K^{-1}$

$m = 1.5kg$

$E_h = cm\Delta T$

$E_h = 4180 \times 1.5 \times 95$

$E_h = 595{,}650\ J = 595.65kJ$

$E_h =$ **600 kJ** to two significant figures

2. Calculate the heat energy required to heat 1.75kg of water from a temperature of 18°C until it begins to boil.

   $\Delta T = t_2 - t_1 = 100 - 18 = 82°C$

   $c = 4180 Jkg^{-1}K^{-1}$

   $m = 1.75kg$

   $E_h = cm\Delta T$

   $E_h = 4180 \times 1.75 \times 82$

   $E_h = 599{,}830\ J = 599.83kJ$

   $E_h =$ **600 kJ** to two significant figures

3. 57kJ of thermal (heat) energy is supplied to 1.7kg of oil having a specific heat capacity of 2.7 kJ/kgK. If the initial temperature of the oil is 3°C, what will be its final temperature?

   $E_h = 57kJ = 57{,}000\ J$

   $t_1 = 3°C$

   $c = 2.7\ kJkg^{-1}K^{-1} = 2{,}700\ Jkg^{-1}K^{-1}$

   $m = 1.7kg$

   $E_h = cm\Delta T$

   $\Delta T = E_h / (cm)$

   $\Delta T = 57{,}000 / (2700 \times 1.7)$

   $\Delta T = 12.4183$

   $\Delta T = t_2 - t_1$

   $t_2 = \Delta T + t_1$

   $t_2 = 12.4183 + 3$

   $t_2 = 15.4183°C =$ **15°C** to 2 significant figures